Reg.No :.....

Name :.....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
SECOND SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

MG2MDCECT100 -IoT based smart farming

Duration: 1 hours **Maximum Marks: 35**

**Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)*

Students should attempt at least one question from each course outcome to enhance their overall *outcome attainability.*

Part A
Multiple Choice Questions
Answer any 35 questions
Each question carries 1 mark

Write complete answers of your choice in the answer paper against each question number

1	What does IoT stand for? a) Internet of Thingsb) Integration of Technology c) Intelligent Online Trackingd) Internet of Technology	K	1
2	Which statement best defines the Internet of Things (IoT)? a) A collection of unrelated devices without communication b) A system where devices collect and exchange data via the internet c) A type of web browser d) A new social media platform	U	1
3	Who is credited with coining the term "Internet of Things"? a) Tim Berners-Leeb) Kevin Ashton c) Elon Muskd) Bill Gates	K	1
4	What is a key characteristic of IoT devices? a) They must always require human intervention to function b) They collect, share, and analyze data automatically c) They work only with Bluetooth connections d) They are limited to home automation applications	K	1
5	How does IoT help in precision farming? a) By collecting real-time data on soil moisture, weather, and crop health b) By replacing human farmers entirely c) By automating only the sale of crops d) By eliminating the need for fertilizers	R	1
6	What role do sensors play in IoT? a) They store data permanently	K	1

	b) They collect real-time data from the environment c) They act as display screens for IoT devices d) They provide only power to IoT systems		
7	What is the primary function of a microcontroller in an IoT system? a) To provide power to sensors b) To process sensor data and control connected devices c) To function as a storage device d) To act as a wireless communication module	U	1
8	Which feature of microcontrollers makes them ideal for IoT applications? a) High power consumptionb) Large display screens c) Low power consumption and real-time processing capabilities d) Ability to run complex video games	U	1
9	What is the primary function of a sensor in IoT? a) Generate electrical powerb) Collect data from the environment c) Store data permanentlyd) Control mechanical devices	U	1
10	What is the first step in IoT data collection? a) Processingb) Sensing c) Transmissiond) Storage	K	1
11	How can soil moisture sensors help in farming? a) Reduce water wastage b) Reduce soil erosion c) Decrease plant growthd) Increase pesticide use	U	1
12	Which of the following is a commonly used temperature sensor? a) LM35 b) PIR sensor c) Soil moisture sensord) Gas sensor	K	1
13	What do humidity sensors measure? a) Water vapor in the air b) Soil moisture c) Wind speedd) Pressure	U	1
14	Which of the following is NOT an example of an actuator? a) Servo motorb) Relay c) LEDd) Microcontroller	U	1
15	What type of actuator is used in automated irrigation? a) Solenoid valve b) Speaker c) Light sensord) Thermistor	K	1
16	What type of actuator is commonly used in robotic arms? a) Stepper motorb) Ultrasonic sensor c) Thermistord) Light sensor	K	1
17	What does IoT stand for? a) Internet of Thingsb) Internet of Technology c) Integration of Thingsd) Internet of Tasks	K	1
18	Which of the following is a communication protocol commonly used in IoT networks? a) HTTPb) Zigbee c) USBd) HDMI	K	1
19	What is Zigbee primarily used for? a) High-speed internet communication	U	1

	b) Long-distance communication c) Short-range wireless communication in IoT devices d) Satellite communication		
20	What is the main advantage of LoRa technology? a) High data transfer speeds b) Long-range communication with low power consumption c) Compatibility with Bluetooth devices d) Low cost for short-distance communication	U	1
21	Which of the following is essential for plant growth? a) Soil, water, and sunlightb) Soil, water, and plastic c) Soil, water, and metald) Soil, water, and sand	K	2
22	What is the first stage of a plant's life cycle? a) Floweringb) Germination c) Pollinationd) Fruiting	K	2
23	During which stage does a plant develop its roots and first leaves? a) Flowering stageb) Germination stage c) Pollination staged) Fruiting stage	K	2
24	Which of the following is NOT a factor influencing plant health? a) Soil qualityb) Amount of sunlight c) Type of pot usedd) Water availability	K	2
25	What is the primary factor that affects plant yield in agriculture? a) The color of the plants b) Availability of nutrients, water, and sunlight c) The height of the plants d) The number of insects present in the soil	U	2
26	Which of the following is a major challenge in traditional farming? a) Lack of natural fertilizers b) Over-reliance on automated machinery c) Dependency on weather conditions and soil quality d) Excessive production of crops	U	2
27	Why is water conservation important in agriculture? a) It helps increase the cost of farming b) It prevents soil erosion and ensures sustainability c) It reduces the amount of sunlight crops receive d) It eliminates the need for irrigation	U	2
28	Which of the following is a common problem faced in irrigation? a) Excessive rainfall throughout the year b) Unequal water distribution and water scarcity c) The inability of soil to absorb nutrients d) Overuse of artificial lighting in crop production	U	2
29	What is a major disadvantage of excessive pesticide usage in farming? a) It makes plants grow faster b) It leads to soil and water contamination c) It completely removes all pests from crops d) It improves the taste of fruits and vegetables	U	2
30	How does modern agriculture impact the environment? a) It always improves soil fertility b) It can lead to deforestation, water pollution, and biodiversity loss	U	2

c) It eliminates the need for fertilizers

d) It reduces global temperatures

31	Which of the following is a significant impact of climate change on agriculture? a) More consistent rainfall throughout the year b) Increased crop yields in all regions c) Unpredictable weather patterns affecting crop growth d) Permanent elimination of plant diseases	U	2
32	Which modern farming technique helps maximize plant growth with minimal soil use? a) Traditional plowing b) Hydroponics and aeroponics c) Burning of crop residues d) Manual seed planting	U	2
33	In vertical farming, which of the following is the main advantage of growing crops in stacked layers? a) It reduces the need for irrigation. b) It maximizes land use and allows for year-round crop production. c) It eliminates the need for artificial lighting. d) It increases the size of the crops produced.	U	2
34	In a hydroponic system, what is the primary function of the nutrient solution? a) To provide water for the plants. b) To deliver essential nutrients directly to the plant roots without soil. c) To stabilize the temperature of the growing environment. d) To replace the need for artificial lighting.	U	2
35	In an aquaponic system, how do fish contribute to plant growth? a) Fish waste provides essential nutrients for the plants. b) Fish help pollinate the plants. c) Fish absorb excess water from the plants. d) Fish regulate the temperature of the plants.	U	2
36	How does the use of data in farming improve the overall efficiency of agricultural practices? a) It increases the cost of farming by requiring more machinery. b) It allows farmers to monitor and optimize crop health, soil conditions, and resource usage. c) It completely replaces traditional farming methods. d) It eliminates the need for physical labor in farming.	U	2
37	In precision agriculture, why is the use of data considered essential for enhancing crop yield? a) It helps farmers apply water and fertilizers in precise amounts, reducing waste. b) It helps farmers decide when to harvest without considering crop health. c) It ensures that crops are watered uniformly across the entire field. d) It replaces traditional farming machinery with fully automated robots.	U	2
38	Which of the following is a common method of data collection in IoT-based smart farming? a) Manual inspections by farmers every few weeks.	U	2

b) Wireless sensors for monitoring soil moisture, temperature, and other environmental factors.

c) Using drones exclusively for aerial surveillance.

d) Collecting data from weather reports alone.

39	In an IoT-based smart farming system, why is cloud storage commonly used for storing farming data? a) It allows data to be stored locally on each IoT device for faster retrieval. b) It enables centralized data storage that can be accessed remotely for analysis. c) It eliminates the need for data analytics in farming. d) It provides an unlimited supply of water for the crops.	U	2
40	What is the primary purpose of data analytics in IoT-based smart farming? a) To reduce the need for human labor in farming operations. b) To analyze environmental data and improve decision-making for farming practices. c) To replace all physical farming equipment with automated systems. d) To generate financial reports for farm owners.	U	2

1 * 35 = 35